Results First Draft

Zachary Serby

October 29, 2026

Summary Statistics

TABLE 1—SUMMARY STATISTICS, 2009 DROUGHT DID SAMPLES

	Panel A: WAZ, age ≤10 in 2005 and 2009					
	Mean	Std. dev.	Min.	Median	Max.	Obs.
Outcomes (MxFLS-2, 2005)						
Weight-for-age z-score (WAZ)	-0.32	1.41	-5.07	-0.37	4.51	1,084
Binary DID treatment (CHIRPS)						
2009 drought (indicator)	0.39	0.49	0.00	0.00	1.00	1,084
2005 drought (indicator)	0.64	0.48	0.00	1.00	1.00	1,084
2006 drought (indicator)	0.06	0.24	0.00	0.00	1.00	1,084
2007 drought (indicator)	0.19	0.40	0.00	0.00	1.00	1,084
2008 drought (indicator)	0.19	0.39	0.00	0.00	1.00	1,084
Continuous DID treatment (CHIRPS)						
2009 rainfall deficit (SD)	0.76	0.72	0.00	0.55	2.51	1,084
Any 2009 rainfall deficit (indicator)	0.77	0.42	0.00	1.00	1.00	1,084
Remittances (MxFLS-1, 2002)						
Household received remittances, proxy (indicator)	0.40	0.49	0.00	0.00	1.00	725
Household (MxFLS-1, 2002)						
Dwelling has electricity (indicator)	0.98	0.15	0.00	1.00	1.00	818
Good floor quality (indicator)	0.83	0.37	0.00	1.00	1.00	818
Good roof quality (indicator)	0.82	0.38	0.00	1.00	1.00	818
Enough ventilation (indicator)	0.56	0.50	0.00	1.00	1.00	812
Sleeping rooms per household member	0.41	0.24	0.11	0.33	2.33	817
Ln (monthly expenditure per capita, pesos), proxy	6.56	0.86	4.21	6.57	11.43	803
Rural locality (indicator)	0.43	0.50	0.00	0.00	1.00	818
Child and mother (MxFLS-2, 2005)						
Age (years, not in regression)	2.43	1.69	0.00	2.00	6.00	1,084
Male (indicator)	0.51	0.50	0.00	1.00	1.00	1,084
Fully vaccinated (indicator)	0.36	0.48	0.00	0.00	1.00	1,029
Mother's height (centimeters)	154.69	6.85	136.00	154.50	182.70	1,033
Mother's years of schooling	7.53	3.42	0.00	8.00	18.00	1,047
Mother's age at birth (years)	26.00	6.49	12.00	25.00	59.00	1,055
Mother employed (indicator)	0.24	0.42	0.00	0.00	1.00	1,046
Father and birth (MxFLS-1 and MxFLS-2)						
Father's height (centimeters)	166.73	7.16	144.00	166.70	192.00	713
Father's years of schooling	7.72	3.64	0.00	9.00	18.00	872
Father's age at birth (years)	29.44	7.74	13.00	28.00	75.00	895
Father employed (indicator)	0.96	0.20	0.00	1.00	1.00	872
Birth order	2.07	1.58	1.00	2.00	13.00	617
Oldest child (indicator)	0.48	0.50	0.00	0.00	1.00	617
Birth weight (kilograms)	3.24	0.56	1.55	3.20	5.50	767
Delivered at home (indicator)	0.07	0.26	0.00	0.00	1.00	818
Weeks predominantly breastfed	32.07	29.15	0.00	26.09	206.00	790
Iron during pregnancy (indicator)	0.78	0.41	0.00	1.00	1.00	799
Premature, mother reported (indicator)	0.18	0.38	0.00	0.00	1.00	796

	Panel B: HAZ and BMIZ, age ≤15 in 2005 and 2009					
	Mean	Std. dev.	Min.	Median	Max.	Obs.
Outcomes (MxFLS-2, 2005)						
Height-for-age z-score (HAZ)	-0.60	1.48	-5.46	-0.64	5.65	2,436
BMI-for-age z-score (BMIZ)	0.36	1.43	-4.95	0.30	4.97	2,393
Binary DID treatment (CHIRPS)						
2009 drought (indicator)	0.41	0.49	0.00	0.00	1.00	2,436
2005 drought (indicator)	0.62	0.49	0.00	1.00	1.00	2,436
2006 drought (indicator)	0.07	0.25	0.00	0.00	1.00	2,436
2007 drought (indicator)	0.20	0.40	0.00	0.00	1.00	2,436
2008 drought (indicator)	0.19	0.39	0.00	0.00	1.00	2,436
Continuous DID treatment (CHIRPS)						
2009 rainfall deficit (SD)	0.79	0.73	0.00	0.60	2.51	2,436
Any 2009 rainfall deficit (indicator)	0.79	0.41	0.00	1.00	1.00	2,436
Remittances (MxFLS-1, 2002)						
Household received remittances, proxy (indicator)	0.36	0.48	0.00	0.00	1.00	1,849
Household (MxFLS-1, 2002)						
Dwelling has electricity (indicator)	0.98	0.15	0.00	1.00	1.00	2,035
Good floor quality (indicator)	0.83	0.38	0.00	1.00	1.00	2,035
Good roof quality (indicator)	0.81	0.39	0.00	1.00	1.00	2,035
Enough ventilation (indicator)	0.56	0.50	0.00	1.00	1.00	2,020
Sleeping rooms per household member	0.38	0.22	0.10	0.33	2.33	2,032
Ln (monthly expenditure per capita, pesos), proxy	6.51	0.85	3.75	6.51	12.24	2,005
Rural locality (indicator)	0.45	0.50	0.00	0.00	1.00	2,035
Child and mother (MxFLS-2, 2005)						
Age (years, not in regression)	5.86	3.29	0.00	6.00	15.00	2,436
Male (indicator)	0.50	0.50	0.00	0.00	1.00	2,436
Fully vaccinated (indicator)	0.48	0.50	0.00	0.00	1.00	2,314
Mother's height (centimeters)	154.30	6.99	133.40	154.10	190.00	2,300
Mother's years of schooling	7.17	3.67	0.00	7.00	18.00	2,328
Mother's age at birth (years)	26.41	6.53	12.00	26.00	59.00	2,354
Mother employed (indicator)	0.29	0.45	0.00	0.00	1.00	2,329
Father and birth (MxFLS-1 and MxFLS-2)						
Father's height (centimeters)	166.38	7.13	144.00	166.20	192.00	1,619
Father's years of schooling	7.40	3.87	0.00	8.00	18.00	1,934
Father's age at birth (years)	29.83	7.84	13.00	29.00	75.00	2,004
Father employed (indicator)	0.96	0.20	0.00	1.00	1.00	1,938
Birth order	2.55	1.88	1.00	2.00	13.00	1,734
Oldest child (indicator)	0.36	0.48	0.00	0.00	1.00	1,734
Birth weight (kilograms)	3.25	0.59	0.85	3.20	6.00	1,589
Delivered at home (indicator)	0.11	0.32	0.00	0.00	1.00	1,924
Weeks predominantly breastfed	33.48	33.28	0.00	26.09	260.89	1,706
Iron during pregnancy (indicator)	0.73	0.44	0.00	1.00	1.00	1,718
Premature, mother reported (indicator)	0.14	0.34	0.00	0.00	1.00	1,714

Notes: Sample is children with the outcome measured in MxFLS-2 (2005) and MxFLS-3 (2009), aged 10 or younger (Panel A) or 15 or younger (Panel B) in both rounds, with a 2005 municipality and not measured in MxFLS-3 before November 2009, as the July–October 2009 rainfall deficit had not yet fully occurred. Dropping them from both groups equalizes treated and control measurement timing. Each row is a child, and Obs. counts children. Statistics are at the 2005 baseline, except remittances and household variables (2002). Panel B reports BMIZ for the HAZ sample, as the BMIZ sample is similar and slightly smaller, at 2,383 children. The continuous DID uses these samples, and binary DID Model 1 excludes 6 (Panel A) and 4 (Panel B) children in singleton interview-month cells. Model 3, the main binary DID, adds 2005–2008 droughts and 2002 household and 2005 child and mother controls and includes 700 (WAZ), 1,755 (HAZ), and 1,714 (BMIZ) children. The SPI analysis compares z of -1 or below with z between -1 and 1, excluding 9 (Panel A) and 36 (Panel B) children with z of 1 or above. Drought is annual CHIRPS rainfall in the 2005 municipality below its 1981–2004 20th percentile. Rainfall deficit is max(0, -z), where z is 2009 rainfall in standard deviations from the 1981–2004 municipal mean. Remittances, a proxy, are cash received in the last 12 months from parents, siblings, or children outside the household. Expenditure is in 2002 pesos per person per month, a proxy for household income. Rural is a locality under 2,500 inhabitants (INEGI, 2024).

Table 2—Sample Sizes, Treated and Control Children

	Panel A: Binary DID, 2009 drought	
	Control	Treated
HAZ sample, children aged 15 or younger in 2005 and 2009		
Children, estimation sample	1,442	994
Children, Model 1 (basic DID)	1,439	993
Children, Model 3 (DID with controls)	1,021	734
Children, Model 5 (triple difference)	946	676
In 2002 remittance households	377	206
Households, estimation sample	894	628
2005 municipalities, estimation sample	80	53
WAZ sample, children aged 10 or younger in 2005 and 2009		
Children, estimation sample	662	422
Children, Model 1 (basic DID)	658	420
Children, Model 3 (DID with controls)	422	278
Children, Model 5 (triple difference)	385	249
In 2002 remittance households	160	98
Households, estimation sample	537	344
2005 municipalities, estimation sample	74	49
BMIZ sample, children aged 15 or younger in 2005 and 2009		
Children, estimation sample	1,415	968
Children, Model 1 (basic DID)	1,412	967
Children, Model 3 (DID with controls)	1,000	714
Children, Model 5 (triple difference)	926	661
In 2002 remittance households	369	201
Households, estimation sample	881	611
2005 municipalities, estimation sample	80	53

	Panel B: Continuous DID, any 2009 deficit	
	Control	Treated
HAZ sample, children aged 15 or younger in 2005 and 2009		
Children, continuous DID sample	518	1,918
In 2002 remittance households	111	563
In 2002 non-remittance households	314	861
Households	324	1,198
2005 municipalities	34	99
WAZ sample, children aged 10 or younger in 2005 and 2009		
Children, continuous DID sample	245	839
In 2002 remittance households	49	244
In 2002 non-remittance households	128	304
Households	200	681
2005 municipalities	30	93
BMIZ sample, children aged 15 or younger in 2005 and 2009		
Children, continuous DID sample	507	1,876
In 2002 remittance households	109	549
In 2002 non-remittance households	307	843
Households	321	1,171
2005 municipalities	34	99

	Panel C: SPI classes vs near normal	
	Control	Treated
Moderately dry, $-1.5 < z \leq -1$		
Children, HAZ sample	1,457	441
Children, WAZ sample	677	191
Children, BMIZ sample	1,433	422
2005 municipalities, HAZ sample	81	26
2005 municipalities, WAZ sample	75	23
2005 municipalities, BMIZ sample	81	26
Severely dry or worse, $z \leq -1.5$		
Children, HAZ sample	1,457	502
Children, WAZ sample	677	207
Children, BMIZ sample	1,433	493
2005 municipalities, HAZ sample	81	24
2005 municipalities, WAZ sample	75	23
2005 municipalities, BMIZ sample	81	24
Severely dry, $-2 < z \leq -1.5$		
Children, HAZ sample	1,457	389
Children, WAZ sample	677	161
Children, BMIZ sample	1,433	381
2005 municipalities, HAZ sample	81	18
2005 municipalities, WAZ sample	75	18
2005 municipalities, BMIZ sample	81	18
Extremely dry, $z \leq -2$		
Children, HAZ sample	1,457	113
Children, WAZ sample	677	46
Children, BMIZ sample	1,433	112
2005 municipalities, HAZ sample	81	6
2005 municipalities, WAZ sample	75	5
2005 municipalities, BMIZ sample	81	6

Notes: Counts are children, households, or 2005 municipalities in each regression sample, at the 2005 baseline. The estimation samples are children with the outcome measured in both MxFLS-2 (2005) and MxFLS-3 (2009), aged 15 or younger (HAZ and BMIZ) or 10 or younger (WAZ) in both rounds. Children have a 2005 municipality and were not measured in MxFLS-3 before November 2009, as the July–October 2009 rainfall deficit had not yet fully occurred. Dropping them from both groups equalizes treated and control measurement timing. In Panel A, treated children's 2005 municipality had a drought in 2009, with annual CHIRPS rainfall below its 1981–2004 20th percentile. Model 1 drops children alone in their interview-month cell, and the continuous DID, with no interview-month fixed effects, uses the full sample. Model 3 adds the 2005–2008 droughts and 2002 household and 2005 child and mother controls, and keeps children with all of them observed. Model 5 adds 2002 remittances times post and the drought times post times 2002 remittances, and keeps children with 2002 remittances observed, so Model 5 children not in remittance households are the Model 5 count minus the remittance row. In Panel B, treated children's 2005 municipality had 2009 rainfall below its 1981–2004 mean (z below 0), where z is 2009 rainfall in standard deviations from that mean. These are the continuous DID samples, and the two remittance rows are the groups of the remittance split. They do not add up to the total, as children with missing 2002 remittances are in neither row. Panel C compares each SPI class with near normal (z between -1 and 1) in the continuous DID. As McKee et al. (1993) note, drought intensity is arbitrarily defined for values of the SPI with these categories. Children in wet municipalities (z of 1 or above) are not in the SPI analysis. Remittances are cash received in the last 12 months from parents, siblings, or children living outside the household, from MxFLS-1 (2002).

Table 3—Sample Attrition Across Waves

	Number of children (N)		
	MxFLS-1 (2002)	MxFLS-2 (2005)	MxFLS-3 (2009)
HAZ sample, children under 16			
DID (2005 and 2009)		3,582	2,436
Event studies (all three waves)	2,723	1,856	1,300
WAZ sample, children under 11			
DID (2005 and 2009)		1,895	1,084
Event studies (all three waves)	962	609	275
BMIZ sample, children under 16			
DID (2005 and 2009)		3,511	2,383
Event studies (all three waves)	2,569	1,719	1,203

Notes: Counts of children in each wave. The DID only uses MxFLS-2 (2005) and MxFLS-3 (2009), and the event studies use all three waves. The continuous DID uses the same samples. Each row starts from the children who meet the regression sample rules in its first wave and could stay in the sample, not from all children measured. They are under the age cutoff in every wave, 15 or younger (HAZ and BMIZ) or 10 or younger (WAZ), have a municipality, and were not measured in MxFLS-3 before November 2009, as the July–October 2009 rainfall deficit had not yet fully occurred. Dropping them from both groups equalizes treated and control measurement timing. Later columns count the children still in the sample, so the drop across columns is attrition. This is not the same as the children the regressions drop as singletons, with only one wave in the sample, as these also include children who aged out or are only in a later wave. The DID 2009 count is the estimation sample, and the event study 2009 count is the children in all three waves. For children not measured in a later wave, the age then comes from their birth date and their household's MxFLS-3 measurement date, as MxFLS-3 interviews ran over several years. Children not in MxFLS-3, or whose birth date does not agree with their reported age, get their last measured age plus the median age gap between the waves, 4 years from 2005 to 2009 and 8 years from 2002 to 2009. This is an estimate, but 92 and 86 percent of these gaps are within a year of the median among children measured in both waves, and a guess off by a year only matters for children near the age cutoff, who could be counted as attrition when they aged out or the other way around. So those who would have aged out anyway are not counted as attrition.

Regression Results

Table 4—2009 Drought and Height-for-Age, Binary DID

	Height-for-age z-score (HAZ)				
	(1)	(2)	(3)	(4)	(5)
Drought × post	0.003	0.049	0.013	0.126	-0.043
	(0.094)	(0.100)	(0.110)	(0.164)	(0.117)
Drought × post × 2002 remittances					-0.093
					(0.143)
2002 remittances × post					-0.188*
					(0.097)
2005–2008 droughts × post	N	Y	Y	Y	Y
2002 household controls × post	N	N	Y	Y	Y
2005 child and mother controls × post	N	N	Y	Y	Y
2005 father and birth controls × post	N	N	N	Y	N
Child FE	Y	Y	Y	Y	Y
Interview year-month FE	Y	Y	Y	Y	Y
Children	2,432	2,432	1,755	734	1,622

Notes: This table reports the effect of the 2009 drought on the WHO height-for-age z-score (HAZ) with the binary DID. Regressions are at the child-wave level. The sample is children with the outcome measured in both MxFLS-2 (2005) and MxFLS-3 (2009), aged 15 or younger in both rounds. Children have a 2005 municipality and were not measured in MxFLS-3 before November 2009, as the July–October 2009 rainfall deficit had not yet fully occurred. Dropping them from both groups equalizes treated and control measurement timing. Drought equals 1 if annual CHIRPS rainfall in 2009 in the child's 2005 municipality was below its 1981–2004 20th percentile, and post equals 1 in MxFLS-3. Remittances equal 1 if the household received cash in the last 12 months from parents, siblings, or children living outside the household in MxFLS-1 (2002), a proxy for remittances. Cols. 1–4 report the average drought effect. Col. 1 is the basic DID, Col. 2 adds the 2005–2008 droughts, and Col. 3 adds 2002 household controls (electricity, floor, roof, ventilation, sleeping rooms per person, log expenditure per capita, rural) and 2005 child and mother controls (child sex and full vaccination, mother's height, schooling, age at birth and employment). Col. 4 adds the 2005 father and birth controls (father's height, schooling, age at birth and employment, birth order, oldest child, birth weight, delivered at home, weeks predominantly breastfed, iron in pregnancy, premature), which are missing for many children, often because the father migrated. All controls are interacted with post. Col. 5 is the triple difference: it keeps the controls of Col. 3 and adds 2002 remittances × post and drought × post × 2002 remittances, the difference in the drought effect for children in 2002 remittance households. Rural is a locality with fewer than 2,500 inhabitants (INEGI, 2024). All regressions include child and interview year-month fixed effects, children in singleton cells are dropped, and standard errors are clustered by 2005 municipality, where the drought is assigned. $^{*}p < .10$, $^{**}p < .05$, $^{***}p < .01$.

Table 5—2009 Drought and Weight-for-Age, Binary DID

	Weight-for-age z-score (WAZ)				
	(1)	(2)	(3)	(4)	(5)
Drought × post	-0.162*	-0.150	-0.321***	-0.247	-0.266**
	(0.096)	(0.107)	(0.115)	(0.194)	(0.130)
Drought × post × 2002 remittances					-0.194
					(0.213)
2002 remittances × post					0.025
					(0.118)
2005–2008 droughts × post	N	Y	Y	Y	Y
2002 household controls × post	N	N	Y	Y	Y
2005 child and mother controls × post	N	N	Y	Y	Y
2005 father and birth controls × post	N	N	N	Y	N
Child FE	Y	Y	Y	Y	Y
Interview year-month FE	Y	Y	Y	Y	Y
Children	1,078	1,078	700	224	634

Notes: This table reports the effect of the 2009 drought on the WHO weight-for-age z-score (WAZ) with the binary DID. Regressions are at the child-wave level. The sample is children with the outcome measured in both MxFLS-2 (2005) and MxFLS-3 (2009), aged 10 or younger in both rounds. Children have a 2005 municipality and were not measured in MxFLS-3 before November 2009, as the July–October 2009 rainfall deficit had not yet fully occurred. Dropping them from both groups equalizes treated and control measurement timing. Drought equals 1 if annual CHIRPS rainfall in 2009 in the child's 2005 municipality was below its 1981–2004 20th percentile, and post equals 1 in MxFLS-3. Remittances equal 1 if the household received cash in the last 12 months from parents, siblings, or children living outside the household in MxFLS-1 (2002), a proxy for remittances. Cols. 1–4 report the average drought effect. Col. 1 is the basic DID, Col. 2 adds the 2005–2008 droughts, and Col. 3 adds 2002 household controls (electricity, floor, roof, ventilation, sleeping rooms per person, log expenditure per capita, rural) and 2005 child and mother controls (child sex and full vaccination, mother's height, schooling, age at birth and employment). Col. 4 adds the 2005 father and birth controls (father's height, schooling, age at birth and employment, birth order, oldest child, birth weight, delivered at home, weeks predominantly breastfed, iron in pregnancy, premature), which are missing for many children, often because the father migrated. All controls are interacted with post. Col. 5 is the triple difference: it keeps the controls of Col. 3 and adds 2002 remittances × post and drought × post × 2002 remittances, the difference in the drought effect for children in 2002 remittance households. Rural is a locality with fewer than 2,500 inhabitants (INEGI, 2024). All regressions include child and interview year-month fixed effects, children in singleton cells are dropped, and standard errors are clustered by 2005 municipality, where the drought is assigned. $^{*}p < .10$, $^{**}p < .05$, $^{***}p < .01$.

Table 6—2009 Drought and BMI-for-Age, Binary DID

	BMI-for-age z-score (BMIZ)				
	(1)	(2)	(3)	(4)	(5)
Drought × post	-0.112	-0.094	-0.167*	-0.068	-0.149
	(0.073)	(0.077)	(0.086)	(0.105)	(0.104)
Drought × post × 2002 remittances					-0.006
					(0.211)
2002 remittances × post					0.032
					(0.074)
2005–2008 droughts × post	N	Y	Y	Y	Y
2002 household controls × post	N	N	Y	Y	Y
2005 child and mother controls × post	N	N	Y	Y	Y
2005 father and birth controls × post	N	N	N	Y	N
Child FE	Y	Y	Y	Y	Y
Interview year-month FE	Y	Y	Y	Y	Y
Children	2,379	2,379	1,714	712	1,587

Notes: This table reports the effect of the 2009 drought on the WHO BMI-for-age z-score (BMIZ) with the binary DID. Regressions are at the child-wave level. The sample is children with the outcome measured in both MxFLS-2 (2005) and MxFLS-3 (2009), aged 15 or younger in both rounds. Children have a 2005 municipality and were not measured in MxFLS-3 before November 2009, as the July–October 2009 rainfall deficit had not yet fully occurred. Dropping them from both groups equalizes treated and control measurement timing. Drought equals 1 if annual CHIRPS rainfall in 2009 in the child's 2005 municipality was below its 1981–2004 20th percentile, and post equals 1 in MxFLS-3. Remittances equal 1 if the household received cash in the last 12 months from parents, siblings, or children living outside the household in MxFLS-1 (2002), a proxy for remittances. Cols. 1–4 report the average drought effect. Col. 1 is the basic DID, Col. 2 adds the 2005–2008 droughts, and Col. 3 adds 2002 household controls (electricity, floor, roof, ventilation, sleeping rooms per person, log expenditure per capita, rural) and 2005 child and mother controls (child sex and full vaccination, mother's height, schooling, age at birth and employment). Col. 4 adds the 2005 father and birth controls (father's height, schooling, age at birth and employment, birth order, oldest child, birth weight, delivered at home, weeks predominantly breastfed, iron in pregnancy, premature), which are missing for many children, often because the father migrated. All controls are interacted with post. Col. 5 is the triple difference: it keeps the controls of Col. 3 and adds 2002 remittances × post and drought × post × 2002 remittances, the difference in the drought effect for children in 2002 remittance households. Rural is a locality with fewer than 2,500 inhabitants (INEGI, 2024). All regressions include child and interview year-month fixed effects, children in singleton cells are dropped, and standard errors are clustered by 2005 municipality, where the drought is assigned. $^{*}p < .10$, $^{**}p < .05$, $^{***}p < .01$.

Table 7—2009 Drought and Child Anthropometrics, SPI Drought Classes

	HAZ		WAZ		BMIZ	
	(1)	(2)	(3)	(4)	(5)	(6)
Dry, $z \leq -1$	-0.003		-0.206*		-0.127	
	(0.098)		(0.111)		(0.076)	
	[0.974]		[0.067]		[0.106]	
Moderately dry, $-1.5 < z \leq -1$		0.162		-0.080		-0.125
		(0.105)		(0.118)		(0.132)
		[0.156]		[0.521]		[0.456]
Severely dry or worse, $z \leq -1.5$		-0.149		-0.322*		-0.129*
		(0.126)		(0.168)		(0.067)
		[0.296]		[0.050]		[0.070]
P-val: pre-trend 2002–2005, moderately dry		0.161		0.528		0.556
P-val: pre-trend 2002–2005, severely dry or worse		0.868		0.613		0.020
Children	2,400	2,400	1,075	1,075	2,348	2,348
Dry 2005 municipalities	50		46		50	
Near-normal 2005 municipalities	81	81	75	75	81	81

Notes: This table compares children in dry 2005 municipalities with children in near-normal ones. Regressions are at the child level, on the 2005 to 2009 change in the WHO z-score, for the binary DID estimation samples without children in wet municipalities (z of 1 or above). z is 2009 annual CHIRPS rainfall in the child's 2005 municipality in standard deviations from its 1981–2004 mean, grouped with the SPI classes of McKee et al. (1993): near normal is z between -1 and 1, moderately dry between -1.5 and -1, and severely dry or worse -1.5 or below. Cols. 1, 3 and 5 compare dry (z of -1 or below) with near normal, and Cols. 2, 4 and 6 split dry into moderately dry and severely dry or worse. Each class is only compared with near normal, which needs parallel trends, and not with the other class, which would need strong parallel trends (Callaway et al., 2024). Since the contdid package does not yet support discrete doses, the classes are estimated with the discrete-dose regression in equation 4.1 (Callaway et al., 2024). Standard errors clustered by 2005 municipality are in parentheses. Wild cluster bootstrap p-values are in brackets, which tend to be conservative with few treated municipalities (MacKinnon et al., 2023). Stars use the clustered p-values, and where a wild bootstrap p-value is shown both must pass: $^{*}p < .10$, $^{**}p < .05$, $^{***}p < .01$.

Table 8—Continuous DID and Remittances as a Moderator

	HAZ (1)	WAZ (2)	BMIZ (3)
Panel A: Continuous DID, any 2009 rainfall deficit			
Average ATT	-0.206**	-0.036	-0.026
	(0.089)	(0.092)	(0.080)
	[0.040]	[0.696]	[0.760]
Average ACRT	-0.096	0.050	0.015
	(0.167)	(0.139)	(0.083)
Children	2,436	1,084	2,383
Panel B: Average ATT by 2002 remittances			
No 2002 remittances	-0.193**	-0.055	-0.077
	(0.095)	(0.146)	(0.100)
2002 remittances	-0.484***	-0.380**	0.034
	(0.157)	(0.169)	(0.164)
Difference, remittances minus none	-0.291*	-0.325	0.111
	(0.171)	(0.224)	(0.186)
Children	1,849	725	1,808
Panel C: Dry vs near normal by 2002 remittances			
Dry, $z \leq -1$	-0.069	-0.233	-0.163
	(0.113)	(0.182)	(0.107)
	[0.573]	[0.260]	[0.156]
Dry × 2002 remittances	-0.081	-0.115	0.073
	(0.136)	(0.216)	(0.199)
	[0.559]	[0.602]	[0.724]
2002 remittances	-0.066	0.057	0.062
	(0.108)	(0.128)	(0.071)
	[0.553]	[0.664]	[0.392]
Children	1,814	716	1,774

Notes: This table reports the continuous DID of Callaway et al. (2024) and tests whether 2002 remittances moderated the effect of the 2009 drought. Regressions are at the child level, on the 2005 to 2009 change in the WHO z-score, for the binary DID estimation samples. The dose is the 2009 rainfall deficit, max(0, -z), where z is 2009 annual CHIRPS rainfall in the child's 2005 municipality in standard deviations from its 1981–2004 mean. Panel A compares children with any 2009 deficit (z below 0) with children without one, Panel B gives the average ATT separately for children in 2002 remittance and non-remittance households and their difference, and Panel C interacts dry (z of -1 or below) vs near normal with 2002 remittances. Remittances are cash received in the last 12 months from parents, siblings, or children living outside the household in MxFLS-1 (2002), a proxy for remittances. The average ATT needs parallel trends, while the average ACRT, the average slope of the dose response, and comparisons between doses need strong parallel trends (Callaway et al., 2024). Panel A uses the contdid package (Callaway et al., 2024). Panels B and C are regressions on the 2005 to 2009 change, and Panel C uses the discrete-dose regression in equation 4.1 (Callaway et al., 2024). The estimator does not allow controls or fixed effects. Standard errors in parentheses are clustered by 2005 municipality, where the drought is assigned. Because the package only bootstraps at the child level, we coded a multiplier bootstrap of its influence functions that clusters by 2005 municipality, and the other rows use municipality-clustered regression standard errors. Wild cluster bootstrap p-values are in brackets, which tend to be conservative with few treated municipalities (MacKinnon et al., 2023). Stars use the clustered p-values, and where a wild bootstrap p-value is shown both must pass: $^{*}p < .10$, $^{**}p < .05$, $^{***}p < .01$.

FIGURE 1—2009 RAINFALL BY MUNICIPALITY, SPI CLASSES

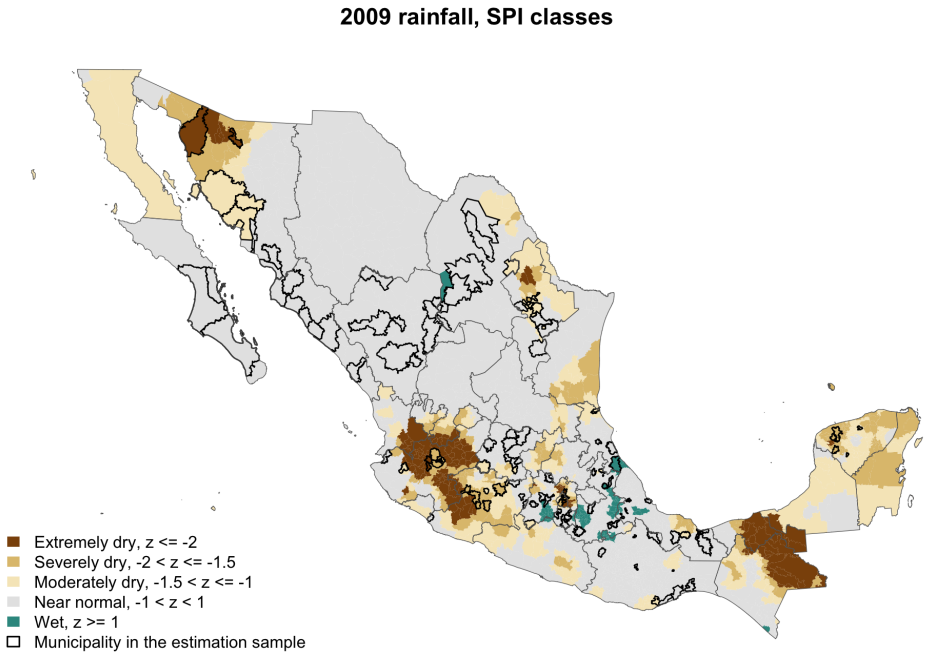


FIGURE 2—2009 DROUGHT BY MUNICIPALITY, BINARY DID TREATMENT

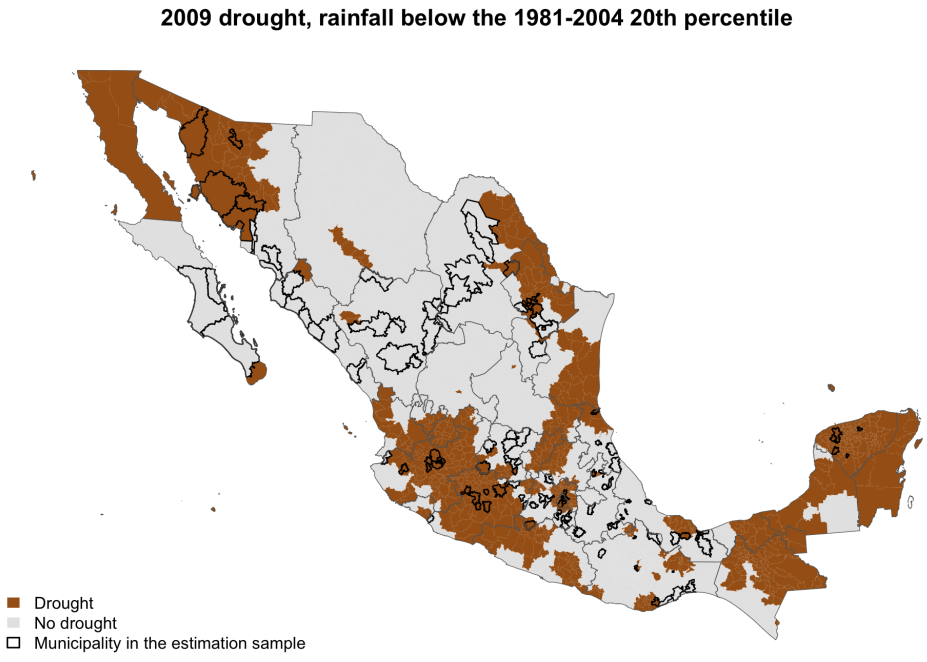


Figure 3—Event Study, Height-for-Age, Binary DID

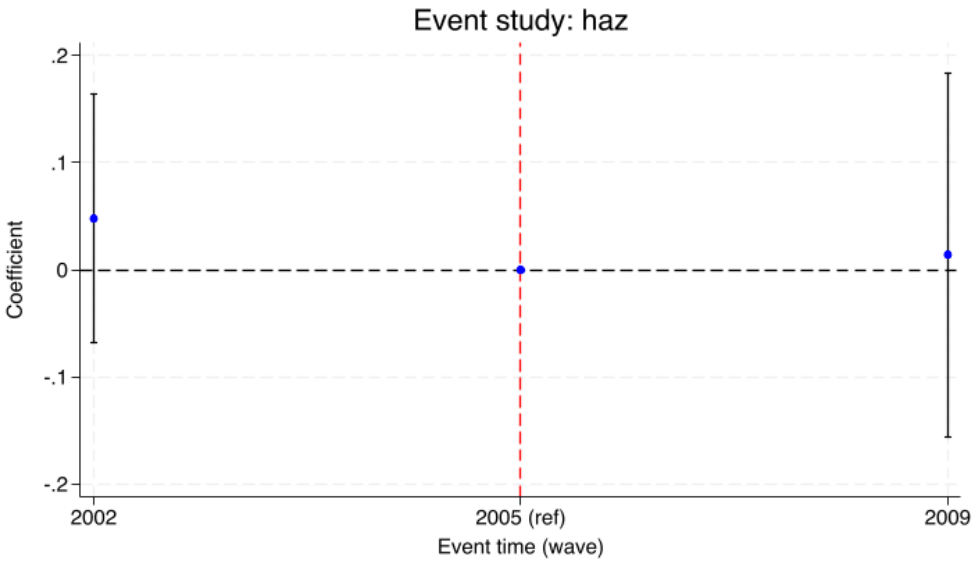


Figure 4—Event Study, Weight-for-Age, Binary DID

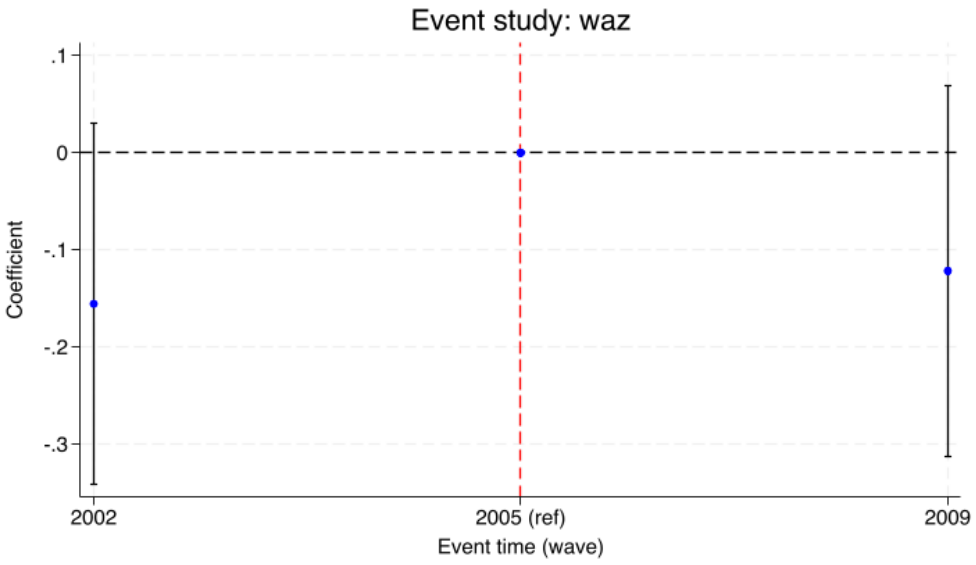


Figure 5—Event Study, BMI-for-Age, Binary DID

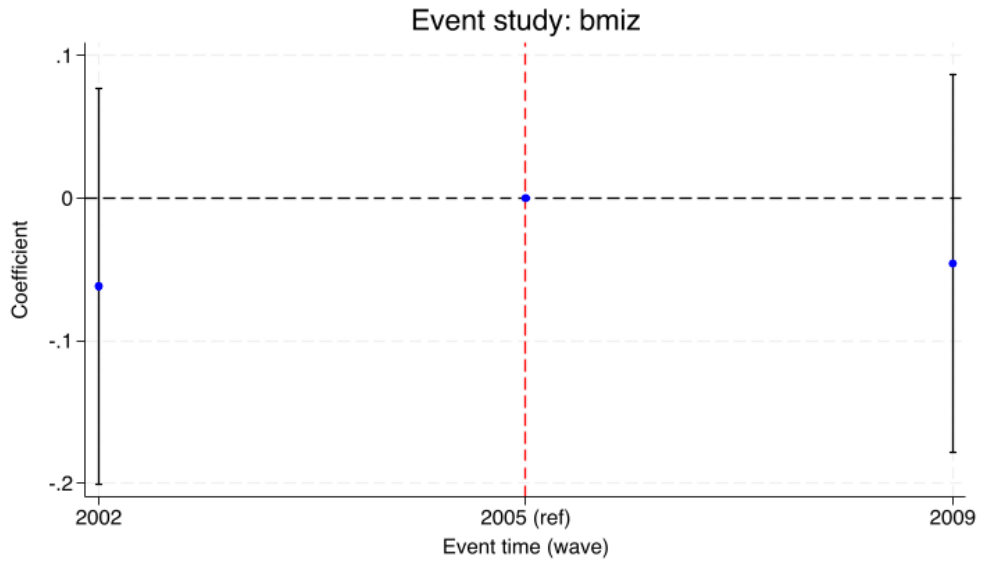


Figure 6—Change in Height-for-Age by 2009 Rainfall Deficit, Continuous DID

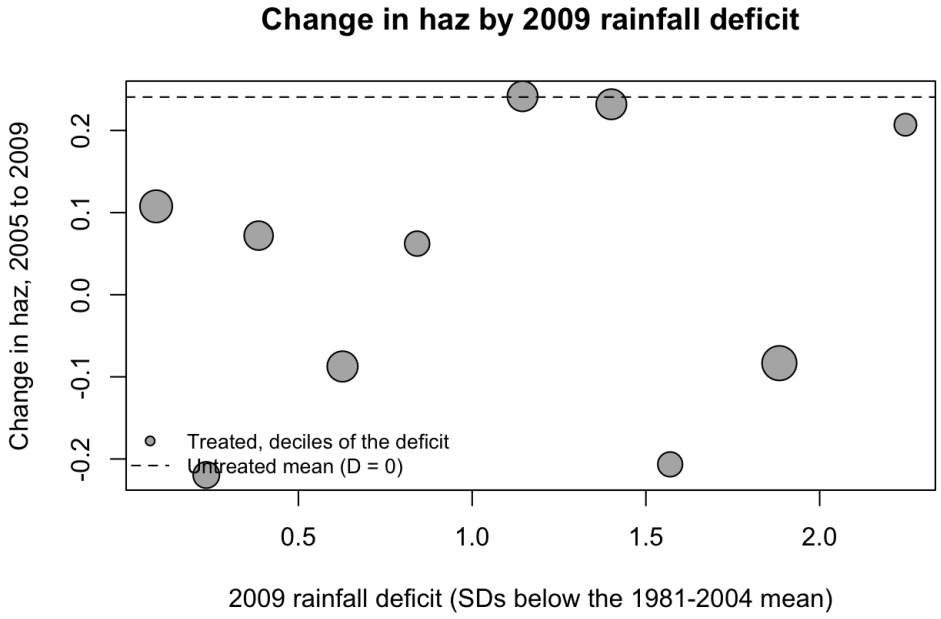


FIGURE 7—CHANGE IN WEIGHT-FOR-AGE BY 2009 RAINFALL DEFICIT, CONTINUOUS DID

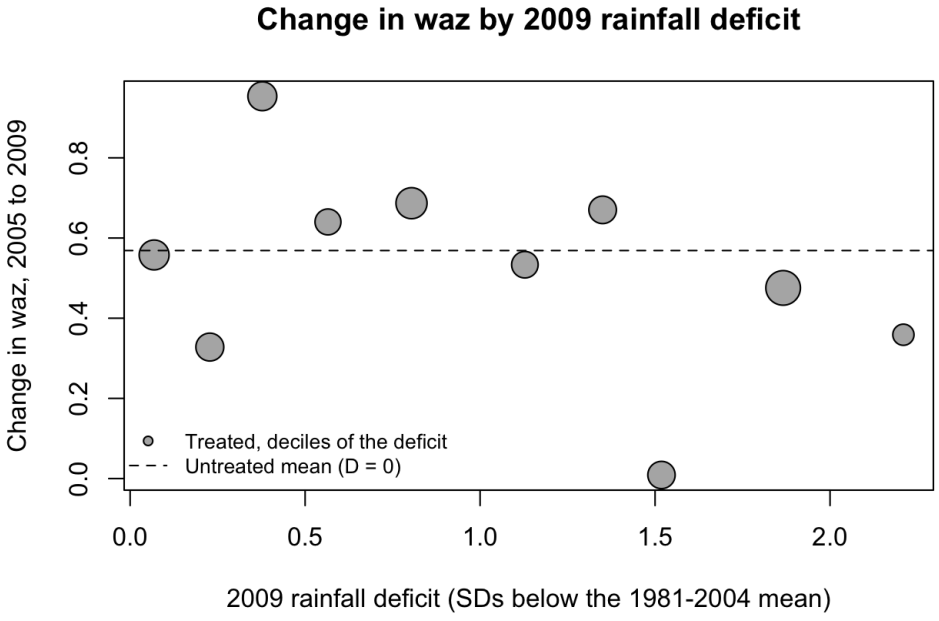
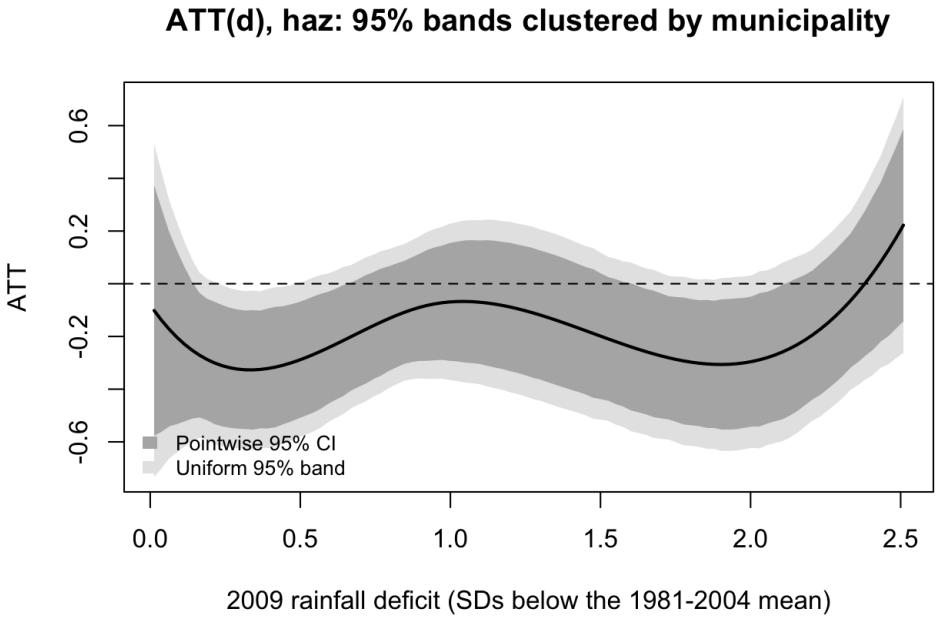


FIGURE 8—ATT BY 2009 RAINFALL DEFICIT, HEIGHT-FOR-AGE, CONTINUOUS DID



Appendix

Table 9—2009 Drought and Height-for-Age, Binary DID, All Three Waves

	Height-for-age z-score (HAZ)				
	(1)	(2)	(3)	(4)	(5)
Drought × post	0.081	0.128	0.138	0.177	0.135
	(0.085)	(0.089)	(0.095)	(0.148)	(0.100)
Drought × post × 2002 remittances					-0.127
					(0.139)
2002 remittances × post					-0.099
					(0.094)
2005–2008 droughts × post	N	Y	Y	Y	Y
2002 household controls × post	N	N	Y	Y	Y
2005 child and mother controls × post	N	N	Y	Y	Y
2005 father and birth controls × post	N	N	N	Y	N
Child FE	Y	Y	Y	Y	Y
Interview year-month FE	Y	Y	Y	Y	Y
Children	1,293	1,293	1,084	559	1,003

Notes: This robustness check keeps only children measured in all three waves (2002, 2005 and 2009), so the event study pre-trend test and the 2005–2009 DID come from the same children, and changes in sample composition, such as attrition or children first measured in 2005, cannot drive the estimates. The table reports the effect of the 2009 drought on the WHO height-for-age z-score (HAZ) with the binary DID for these children, aged 15 or younger in both rounds, with a 2005 municipality and not measured in MxFLS-3 before November 2009, as the July–October 2009 rainfall deficit had not yet fully occurred. Dropping them from both groups equalizes treated and control measurement timing. Columns, variables and controls are as in the main binary DID tables. All regressions include child and interview year-month fixed effects, and standard errors are clustered by 2005 municipality. $^{*}p < .10$, $^{**}p < .05$, $^{***}p < .01$.

Table 10—2009 Drought and Weight-for-Age, Binary DID, All Three Waves

	Weight-for-age z-score (WAZ)				
	(1)	(2)	(3)	(4)	(5)
Drought × post	0.016	-0.068	-0.026	-0.078	0.166
	(0.127)	(0.151)	(0.163)	(0.252)	(0.210)
Drought × post × 2002 remittances					-0.142
					(0.383)
2002 remittances × post					0.008
					(0.273)
2005–2008 droughts × post	N	Y	Y	Y	Y
2002 household controls × post	N	N	Y	Y	Y
2005 child and mother controls × post	N	N	Y	Y	Y
2005 father and birth controls × post	N	N	N	Y	N
Child FE	Y	Y	Y	Y	Y
Interview year-month FE	Y	Y	Y	Y	Y
Children	266	266	203	121	179

Notes: This robustness check keeps only children measured in all three waves (2002, 2005 and 2009), so the event study pre-trend test and the 2005–2009 DID come from the same children, and changes in sample composition, such as attrition or children first measured in 2005, cannot drive the estimates. The table reports the effect of the 2009 drought on the WHO weight-for-age z-score (WAZ) with the binary DID for these children, aged 10 or younger in both rounds, with a 2005 municipality and not measured in MxFLS-3 before November 2009, as the July–October 2009 rainfall deficit had not yet fully occurred. Dropping them from both groups equalizes treated and control measurement timing. Columns, variables and controls are as in the main binary DID tables. All regressions include child and interview year-month fixed effects, and standard errors are clustered by 2005 municipality. $^{*}p < .10$, $^{**}p < .05$, $^{***}p < .01$.

Table 11—2009 Drought and BMI-for-Age, Binary DID, All Three Waves

	BMI-for-age z-score (BMIZ)				
	(1)	(2)	(3)	(4)	(5)
Drought × post	-0.038	0.006	-0.010	-0.069	0.044
	(0.098)	(0.105)	(0.110)	(0.108)	(0.127)
Drought × post × 2002 remittances					-0.092
					(0.235)
2002 remittances × post					0.064
					(0.087)
2005–2008 droughts × post	N	Y	Y	Y	Y
2002 household controls × post	N	N	Y	Y	Y
2005 child and mother controls × post	N	N	Y	Y	Y
2005 father and birth controls × post	N	N	N	Y	N
Child FE	Y	Y	Y	Y	Y
Interview year-month FE	Y	Y	Y	Y	Y
Children	1,196	1,196	1,006	513	933

Notes: This robustness check keeps only children measured in all three waves (2002, 2005 and 2009), so the event study pre-trend test and the 2005–2009 DID come from the same children, and changes in sample composition, such as attrition or children first measured in 2005, cannot drive the estimates. The table reports the effect of the 2009 drought on the WHO BMI-for-age z-score (BMIZ) with the binary DID for these children, aged 15 or younger in both rounds, with a 2005 municipality and not measured in MxFLS-3 before November 2009, as the July–October 2009 rainfall deficit had not yet fully occurred. Dropping them from both groups equalizes treated and control measurement timing. Columns, variables and controls are as in the main binary DID tables. All regressions include child and interview year-month fixed effects, and standard errors are clustered by 2005 municipality. $^{*}p < .10$, $^{**}p < .05$, $^{***}p < .01$.

Table 12—2009 Drought and Height-for-Age, Binary DID, Boys

	Height-for-age z-score (HAZ)				
	(1)	(2)	(3)	(4)	(5)
Drought × post	-0.020	0.066	-0.016	0.258	-0.032
	(0.126)	(0.136)	(0.149)	(0.250)	(0.141)
Drought × post × 2002 remittances					-0.214
					(0.233)
2002 remittances × post					-0.273**
					(0.130)
2005–2008 droughts × post	N	Y	Y	Y	Y
2002 household controls × post	N	N	Y	Y	Y
2005 child and mother controls × post	N	N	Y	Y	Y
2005 father and birth controls × post	N	N	N	Y	N
Child FE	Y	Y	Y	Y	Y
Interview year-month FE	Y	Y	Y	Y	Y
Children	1,201	1,201	869	353	806

Notes: Estimating the effect separately for boys and girls tests for gender disparities in the impact of the drought, as households facing an income shock may protect the nutrition and health of sons and daughters differently. The table reports the effect of the 2009 drought on the WHO height-for-age z-score (HAZ) with the binary DID for boys (2005 roster sex), aged 15 or younger in both rounds, with a 2005 municipality and not measured in MxFLS-3 before November 2009, as the July–October 2009 rainfall deficit had not yet fully occurred. Dropping them from both groups equalizes treated and control measurement timing. Columns, variables and controls are as in the main binary DID tables, without child sex in the controls. All regressions include child and interview year-month fixed effects, and standard errors are clustered by 2005 municipality. $^{*}p < .10$, $^{**}p < .05$, $^{***}p < .01$.

Table 13—2009 Drought and Weight-for-Age, Binary DID, Boys

	Weight-for-age z-score (WAZ)				
	(1)	(2)	(3)	(4)	(5)
Drought × post	-0.168	-0.125	-0.269	-0.426	-0.140
	(0.149)	(0.161)	(0.182)	(0.326)	(0.212)
Drought × post × 2002 remittances					-0.022
					(0.293)
2002 remittances × post					0.008
					(0.179)
2005–2008 droughts × post	N	Y	Y	Y	Y
2002 household controls × post	N	N	Y	Y	Y
2005 child and mother controls × post	N	N	Y	Y	Y
2005 father and birth controls × post	N	N	N	Y	N
Child FE	Y	Y	Y	Y	Y
Interview year-month FE	Y	Y	Y	Y	Y
Children	551	551	367	103	331

Notes: Estimating the effect separately for boys and girls tests for gender disparities in the impact of the drought, as households facing an income shock may protect the nutrition and health of sons and daughters differently. The table reports the effect of the 2009 drought on the WHO weight-for-age z-score (WAZ) with the binary DID for boys (2005 roster sex), aged 10 or younger in both rounds, with a 2005 municipality and not measured in MxFLS-3 before November 2009, as the July–October 2009 rainfall deficit had not yet fully occurred. Dropping them from both groups equalizes treated and control measurement timing. Columns, variables and controls are as in the main binary DID tables, without child sex in the controls. All regressions include child and interview year-month fixed effects, and standard errors are clustered by 2005 municipality. $^{*}p < .10$, $^{**}p < .05$, $^{***}p < .01$.

Table 14—2009 Drought and BMI-for-Age, Binary DID, Boys

	BMI-for-age z-score (BMIZ)				
	(1)	(2)	(3)	(4)	(5)
Drought × post	-0.160	-0.141	-0.214*	-0.189	-0.332**
	(0.101)	(0.103)	(0.116)	(0.149)	(0.150)
Drought × post × 2002 remittances					0.459
					(0.298)
2002 remittances × post					-0.083
					(0.116)
2005–2008 droughts × post	N	Y	Y	Y	Y
2002 household controls × post	N	N	Y	Y	Y
2005 child and mother controls × post	N	N	Y	Y	Y
2005 father and birth controls × post	N	N	N	Y	N
Child FE	Y	Y	Y	Y	Y
Interview year-month FE	Y	Y	Y	Y	Y
Children	1,170	1,170	847	346	787

Notes: Estimating the effect separately for boys and girls tests for gender disparities in the impact of the drought, as households facing an income shock may protect the nutrition and health of sons and daughters differently. The table reports the effect of the 2009 drought on the WHO BMI-for-age z-score (BMIZ) with the binary DID for boys (2005 roster sex), aged 15 or younger in both rounds, with a 2005 municipality and not measured in MxFLS-3 before November 2009, as the July–October 2009 rainfall deficit had not yet fully occurred. Dropping them from both groups equalizes treated and control measurement timing. Columns, variables and controls are as in the main binary DID tables, without child sex in the controls. All regressions include child and interview year-month fixed effects, and standard errors are clustered by 2005 municipality. $^{*}p < .10$, $^{**}p < .05$, $^{***}p < .01$.

Table 15—2009 Drought and Height-for-Age, Binary DID, Girls

	Height-for-age z-score (HAZ)				
	(1)	(2)	(3)	(4)	(5)
Drought × post	0.007	0.015	-0.004	0.035	-0.085
	(0.096)	(0.103)	(0.116)	(0.194)	(0.134)
Drought × post × 2002 remittances					0.002
					(0.182)
2002 remittances × post					-0.171
					(0.122)
2005–2008 droughts × post	N	Y	Y	Y	Y
2002 household controls × post	N	N	Y	Y	Y
2005 child and mother controls × post	N	N	Y	Y	Y
2005 father and birth controls × post	N	N	N	Y	N
Child FE	Y	Y	Y	Y	Y
Interview year-month FE	Y	Y	Y	Y	Y
Children	1,220	1,220	878	377	811

Notes: Estimating the effect separately for boys and girls tests for gender disparities in the impact of the drought, as households facing an income shock may protect the nutrition and health of sons and daughters differently. The table reports the effect of the 2009 drought on the WHO height-for-age z-score (HAZ) with the binary DID for girls (2005 roster sex), aged 15 or younger in both rounds, with a 2005 municipality and not measured in MxFLS-3 before November 2009, as the July–October 2009 rainfall deficit had not yet fully occurred. Dropping them from both groups equalizes treated and control measurement timing. Columns, variables and controls are as in the main binary DID tables, without child sex in the controls. All regressions include child and interview year-month fixed effects, and standard errors are clustered by 2005 municipality. $^{*}p < .10$, $^{**}p < .05$, $^{***}p < .01$.

Table 16—2009 Drought and Weight-for-Age, Binary DID, Girls

	Weight-for-age z-score (WAZ)				
	(1)	(2)	(3)	(4)	(5)
Drought × post	-0.248**	-0.303**	-0.508***	-0.593*	-0.507***
	(0.107)	(0.116)	(0.137)	(0.316)	(0.170)
Drought × post × 2002 remittances					-0.316
					(0.323)
2002 remittances × post					0.056
					(0.189)
2005–2008 droughts × post	N	Y	Y	Y	Y
2002 household controls × post	N	N	Y	Y	Y
2005 child and mother controls × post	N	N	Y	Y	Y
2005 father and birth controls × post	N	N	N	Y	N
Child FE	Y	Y	Y	Y	Y
Interview year-month FE	Y	Y	Y	Y	Y
Children	520	520	324	115	295

Notes: Estimating the effect separately for boys and girls tests for gender disparities in the impact of the drought, as households facing an income shock may protect the nutrition and health of sons and daughters differently. The table reports the effect of the 2009 drought on the WHO weight-for-age z-score (WAZ) with the binary DID for girls (2005 roster sex), aged 10 or younger in both rounds, with a 2005 municipality and not measured in MxFLS-3 before November 2009, as the July–October 2009 rainfall deficit had not yet fully occurred. Dropping them from both groups equalizes treated and control measurement timing. Columns, variables and controls are as in the main binary DID tables, without child sex in the controls. All regressions include child and interview year-month fixed effects, and standard errors are clustered by 2005 municipality. $^{*}p < .10$, $^{**}p < .05$, $^{***}p < .01$.

Table 17—2009 Drought and BMI-for-Age, Binary DID, Girls

	BMI-for-age z-score (BMIZ)				
	(1)	(2)	(3)	(4)	(5)
Drought × post	-0.076	-0.064	-0.151	-0.117	-0.000
	(0.082)	(0.101)	(0.123)	(0.165)	(0.130)
Drought × post × 2002 remittances					-0.489**
					(0.196)
2002 remittances × post					0.133
					(0.097)
2005–2008 droughts × post	N	Y	Y	Y	Y
2002 household controls × post	N	N	Y	Y	Y
2005 child and mother controls × post	N	N	Y	Y	Y
2005 father and birth controls × post	N	N	N	Y	N
Child FE	Y	Y	Y	Y	Y
Interview year-month FE	Y	Y	Y	Y	Y
Children	1,199	1,199	861	364	795

Notes: Estimating the effect separately for boys and girls tests for gender disparities in the impact of the drought, as households facing an income shock may protect the nutrition and health of sons and daughters differently. The table reports the effect of the 2009 drought on the WHO BMI-for-age z-score (BMIZ) with the binary DID for girls (2005 roster sex), aged 15 or younger in both rounds, with a 2005 municipality and not measured in MxFLS-3 before November 2009, as the July–October 2009 rainfall deficit had not yet fully occurred. Dropping them from both groups equalizes treated and control measurement timing. Columns, variables and controls are as in the main binary DID tables, without child sex in the controls. All regressions include child and interview year-month fixed effects, and standard errors are clustered by 2005 municipality. $^{*}p < .10$, $^{**}p < .05$, $^{***}p < .01$.

Table 18—2009 Drought and Height-for-Age, Binary DID, Rural Children

	Height-for-age z-score (HAZ)				
	(1)	(2)	(3)	(4)	(5)
Drought × post	0.079	0.342**	0.443***	0.640***	0.401**
	(0.153)	(0.141)	(0.146)	(0.224)	(0.178)
Drought × post × 2002 remittances					-0.091
					(0.278)
2002 remittances × post					-0.084
					(0.166)
2005–2008 droughts × post	N	Y	Y	Y	Y
2002 household controls × post	N	N	Y	Y	Y
2005 child and mother controls × post	N	N	Y	Y	Y
2005 father and birth controls × post	N	N	N	Y	N
Child FE	Y	Y	Y	Y	Y
Interview year-month FE	Y	Y	Y	Y	Y
Children	1,016	1,016	732	272	714

Notes: Estimating the effect separately for rural and urban children tests whether the drought hits harder where household incomes depend more directly on rain-fed agriculture, while urban households are mainly exposed through food prices. Rural is a locality with fewer than 2,500 inhabitants (INEGI, 2024). The table reports the effect of the 2009 drought on the WHO height-for-age z-score (HAZ) with the binary DID for rural children (2005 household locality), aged 15 or younger in both rounds, with a 2005 municipality and not measured in MxFLS-3 before November 2009, as the July–October 2009 rainfall deficit had not yet fully occurred. Dropping them from both groups equalizes treated and control measurement timing. Columns, variables and controls are as in the main binary DID tables, without rural in the controls. All regressions include child and interview year-month fixed effects, and standard errors are clustered by 2005 municipality. $^{*}p < .10$, $^{**}p < .05$, $^{***}p < .01$.

Table 19—2009 Drought and Weight-for-Age, Binary DID, Rural Children

	Weight-for-age z-score (WAZ)				
	(1)	(2)	(3)	(4)	(5)
Drought × post	-0.204	-0.068	-0.323	-0.457	-0.304
	(0.157)	(0.161)	(0.252)	(0.417)	(0.268)
Drought × post × 2002 remittances					-0.151
					(0.325)
2002 remittances × post					0.108
					(0.219)
2005–2008 droughts × post	N	Y	Y	Y	Y
2002 household controls × post	N	N	Y	Y	Y
2005 child and mother controls × post	N	N	Y	Y	Y
2005 father and birth controls × post	N	N	N	Y	N
Child FE	Y	Y	Y	Y	Y
Interview year-month FE	Y	Y	Y	Y	Y
Children	414	414	265	85	260

Notes: Estimating the effect separately for rural and urban children tests whether the drought hits harder where household incomes depend more directly on rain-fed agriculture, while urban households are mainly exposed through food prices. Rural is a locality with fewer than 2,500 inhabitants (INEGI, 2024). The table reports the effect of the 2009 drought on the WHO weight-for-age z-score (WAZ) with the binary DID for rural children (2005 household locality), aged 10 or younger in both rounds, with a 2005 municipality and not measured in MxFLS-3 before November 2009, as the July–October 2009 rainfall deficit had not yet fully occurred. Dropping them from both groups equalizes treated and control measurement timing. Columns, variables and controls are as in the main binary DID tables, without rural in the controls. All regressions include child and interview year-month fixed effects, and standard errors are clustered by 2005 municipality. $^{*}p < .10$, $^{**}p < .05$, $^{***}p < .01$.

Table 20—2009 Drought and BMI-for-Age, Binary DID, Rural Children

	BMI-for-age z-score (BMIZ)				
	(1)	(2)	(3)	(4)	(5)
Drought × post	-0.054	-0.028	-0.154	-0.227	-0.137
	(0.113)	(0.144)	(0.175)	(0.221)	(0.197)
Drought × post × 2002 remittances					-0.052
					(0.259)
2002 remittances × post					0.097
					(0.113)
2005–2008 droughts × post	N	Y	Y	Y	Y
2002 household controls × post	N	N	Y	Y	Y
2005 child and mother controls × post	N	N	Y	Y	Y
2005 father and birth controls × post	N	N	N	Y	N
Child FE	Y	Y	Y	Y	Y
Interview year-month FE	Y	Y	Y	Y	Y
Children	998	998	719	264	701

Notes: Estimating the effect separately for rural and urban children tests whether the drought hits harder where household incomes depend more directly on rain-fed agriculture, while urban households are mainly exposed through food prices. Rural is a locality with fewer than 2,500 inhabitants (INEGI, 2024). The table reports the effect of the 2009 drought on the WHO BMI-for-age z-score (BMIZ) with the binary DID for rural children (2005 household locality), aged 15 or younger in both rounds, with a 2005 municipality and not measured in MxFLS-3 before November 2009, as the July–October 2009 rainfall deficit had not yet fully occurred. Dropping them from both groups equalizes treated and control measurement timing. Columns, variables and controls are as in the main binary DID tables, without rural in the controls. All regressions include child and interview year-month fixed effects, and standard errors are clustered by 2005 municipality. $^{*}p < .10$, $^{**}p < .05$, $^{***}p < .01$.

Table 21—2009 Drought and Height-for-Age, Binary DID, Urban Children

	Height-for-age z-score (HAZ)				
	(1)	(2)	(3)	(4)	(5)
Drought × post	-0.108	-0.186	-0.277**	-0.302	-0.327**
	(0.110)	(0.130)	(0.127)	(0.197)	(0.140)
Drought × post × 2002 remittances					-0.018
					(0.179)
2002 remittances × post					-0.321**
					(0.127)
2005–2008 droughts × post	N	Y	Y	Y	Y
2002 household controls × post	N	N	Y	Y	Y
2005 child and mother controls × post	N	N	Y	Y	Y
2005 father and birth controls × post	N	N	N	Y	N
Child FE	Y	Y	Y	Y	Y
Interview year-month FE	Y	Y	Y	Y	Y
Children	1,412	1,412	1,020	455	904

Notes: Estimating the effect separately for rural and urban children tests whether the drought hits harder where household incomes depend more directly on rain-fed agriculture, while urban households are mainly exposed through food prices. Rural is a locality with fewer than 2,500 inhabitants (INEGI, 2024). The table reports the effect of the 2009 drought on the WHO height-for-age z-score (HAZ) with the binary DID for urban children (2005 household locality), aged 15 or younger in both rounds, with a 2005 municipality and not measured in MxFLS-3 before November 2009, as the July–October 2009 rainfall deficit had not yet fully occurred. Dropping them from both groups equalizes treated and control measurement timing. Columns, variables and controls are as in the main binary DID tables, without rural in the controls. All regressions include child and interview year-month fixed effects, and standard errors are clustered by 2005 municipality. $^{*}p < .10$, $^{**}p < .05$, $^{***}p < .01$.

Table 22—2009 Drought and Weight-for-Age, Binary DID, Urban Children

	Weight-for-age z-score (WAZ)				
	(1)	(2)	(3)	(4)	(5)
Drought × post	-0.200	-0.351**	-0.508***	-0.385	-0.400**
	(0.127)	(0.138)	(0.142)	(0.288)	(0.159)
Drought × post × 2002 remittances					-0.401
					(0.259)
2002 remittances × post					0.013
					(0.138)
2005–2008 droughts × post	N	Y	Y	Y	Y
2002 household controls × post	N	N	Y	Y	Y
2005 child and mother controls × post	N	N	Y	Y	Y
2005 father and birth controls × post	N	N	N	Y	N
Child FE	Y	Y	Y	Y	Y
Interview year-month FE	Y	Y	Y	Y	Y
Children	654	654	427	137	365

Notes: Estimating the effect separately for rural and urban children tests whether the drought hits harder where household incomes depend more directly on rain-fed agriculture, while urban households are mainly exposed through food prices. Rural is a locality with fewer than 2,500 inhabitants (INEGI, 2024). The table reports the effect of the 2009 drought on the WHO weight-for-age z-score (WAZ) with the binary DID for urban children (2005 household locality), aged 10 or younger in both rounds, with a 2005 municipality and not measured in MxFLS-3 before November 2009, as the July–October 2009 rainfall deficit had not yet fully occurred. Dropping them from both groups equalizes treated and control measurement timing. Columns, variables and controls are as in the main binary DID tables, without rural in the controls. All regressions include child and interview year-month fixed effects, and standard errors are clustered by 2005 municipality. $^{*}p < .10$, $^{**}p < .05$, $^{***}p < .01$.

Table 23—2009 Drought and BMI-for-Age, Binary DID, Urban Children

	BMI-for-age z-score (BMIZ)				
	(1)	(2)	(3)	(4)	(5)
Drought × post	-0.144*	-0.130	-0.208**	-0.006	-0.197
	(0.081)	(0.087)	(0.100)	(0.141)	(0.126)
Drought × post × 2002 remittances					0.021
					(0.314)
2002 remittances × post					0.053
					(0.109)
2005–2008 droughts × post	N	Y	Y	Y	Y
2002 household controls × post	N	N	Y	Y	Y
2005 child and mother controls × post	N	N	Y	Y	Y
2005 father and birth controls × post	N	N	N	Y	N
Child FE	Y	Y	Y	Y	Y
Interview year-month FE	Y	Y	Y	Y	Y
Children	1,377	1,377	992	441	882

Notes: Estimating the effect separately for rural and urban children tests whether the drought hits harder where household incomes depend more directly on rain-fed agriculture, while urban households are mainly exposed through food prices. Rural is a locality with fewer than 2,500 inhabitants (INEGI, 2024). The table reports the effect of the 2009 drought on the WHO BMI-for-age z-score (BMIZ) with the binary DID for urban children (2005 household locality), aged 15 or younger in both rounds, with a 2005 municipality and not measured in MxFLS-3 before November 2009, as the July–October 2009 rainfall deficit had not yet fully occurred. Dropping them from both groups equalizes treated and control measurement timing. Columns, variables and controls are as in the main binary DID tables, without rural in the controls. All regressions include child and interview year-month fixed effects, and standard errors are clustered by 2005 municipality. $^{*}p < .10$, $^{**}p < .05$, $^{***}p < .01$.

Table 24—Event Study, 2009 Drought and Height-for-Age

	Height-for-age z-score (HAZ)	
	(1) Event study	(2) Triple difference
Drought × 2002 wave (lead)	0.048	0.050
	(0.059)	(0.071)
Drought × 2009 wave	0.014	-0.085
	(0.087)	(0.101)
Drought × 2002 remittances × 2002 wave (lead)		-0.082
		(0.112)
Drought × 2002 remittances × 2009 wave		-0.026
		(0.129)
Child FE	Y	Y
Interview year-month FE	Y	Y
P-value, lead = 0	0.422	0.464
2009 drought effect, remittance households		-0.110
SE		0.136
P-value		0.419
Observations (child-waves)	10,399	8,518
Children	4,550	3,690
2005 municipalities (clusters)	143	136

Notes: This table tests for pre-trends before the 2009 drought. Regressions are at the child-wave level and use MxFLS-1 (2002), MxFLS-2 (2005) and MxFLS-3 (2009), with 2005 as the reference wave. The sample is children aged 15 or younger in each wave they are in, with a 2005 municipality, and not measured in MxFLS-3 before November 2009, as the July–October 2009 rainfall deficit had not yet fully occurred. Dropping them from both groups equalizes treated and control measurement timing. Col. 1 regresses the outcome on the drought times the 2002 wave (the lead) and times the 2009 wave. Col. 2 adds the same terms times 2002 remittances, and the 2002 remittance terms on their own (not shown), for children with 2002 remittances observed. The P-value of the lead tests the drought lead in Col. 1 and the triple difference lead in Col. 2. The effect for remittance households is the sum of the two 2009 drought terms. All regressions include child and interview year-month fixed effects, and standard errors are clustered by 2005 municipality. $^{*}p < .10$, $^{**}p < .05$, $^{***}p < .01$.

Table 25—Event Study, 2009 Drought and Weight-for-Age

	Weight-for-age z-score (WAZ)	
	(1) Event study	(2) Triple difference
Drought × 2002 wave (lead)	-0.156	-0.063
	(0.095)	(0.100)
Drought × 2009 wave	-0.122	-0.160
	(0.097)	(0.139)
Drought × 2002 remittances × 2002 wave (lead)		-0.303*
		(0.157)
Drought × 2002 remittances × 2009 wave		-0.113
		(0.186)
Child FE	Y	Y
Interview year-month FE	Y	Y
P-value, lead = 0	0.103	0.056
2009 drought effect, remittance households		-0.273
SE		0.160
P-value		0.090
Observations (child-waves)	4,959	3,848
Children	2,342	1,817
2005 municipalities (clusters)	137	127

Notes: This table tests for pre-trends before the 2009 drought. Regressions are at the child-wave level and use MxFLS-1 (2002), MxFLS-2 (2005) and MxFLS-3 (2009), with 2005 as the reference wave. The sample is children aged 10 or younger in each wave they are in, with a 2005 municipality, and not measured in MxFLS-3 before November 2009, as the July–October 2009 rainfall deficit had not yet fully occurred. Dropping them from both groups equalizes treated and control measurement timing. Col. 1 regresses the outcome on the drought times the 2002 wave (the lead) and times the 2009 wave. Col. 2 adds the same terms times 2002 remittances, and the 2002 remittance terms on their own (not shown), for children with 2002 remittances observed. The P-value of the lead tests the drought lead in Col. 1 and the triple difference lead in Col. 2. The effect for remittance households is the sum of the two 2009 drought terms. All regressions include child and interview year-month fixed effects, and standard errors are clustered by 2005 municipality. $^{*}p < .10$, $^{**}p < .05$, $^{***}p < .01$.

Table 26—Event Study, 2009 Drought and BMI-for-Age

	BMI-for-age z-score (BMIZ)	
	(1) Event study	(2) Triple difference
Drought × 2002 wave (lead)	-0.062	-0.054
	(0.071)	(0.088)
Drought × 2009 wave	-0.046	-0.075
	(0.068)	(0.099)
Drought × 2002 remittances × 2002 wave (lead)		-0.004
		(0.129)
Drought × 2002 remittances × 2009 wave		0.061
		(0.192)
Child FE	Y	Y
Interview year-month FE	Y	Y
P-value, lead = 0	0.385	0.973
2009 drought effect, remittance households		-0.014
SE		0.142
P-value		0.922
Observations (child-waves)	9,978	8,163
Children	4,388	3,554
2005 municipalities (clusters)	143	136

Notes: This table tests for pre-trends before the 2009 drought. Regressions are at the child-wave level and use MxFLS-1 (2002), MxFLS-2 (2005) and MxFLS-3 (2009), with 2005 as the reference wave. The sample is children aged 15 or younger in each wave they are in, with a 2005 municipality, and not measured in MxFLS-3 before November 2009, as the July–October 2009 rainfall deficit had not yet fully occurred. Dropping them from both groups equalizes treated and control measurement timing. Col. 1 regresses the outcome on the drought times the 2002 wave (the lead) and times the 2009 wave. Col. 2 adds the same terms times 2002 remittances, and the 2002 remittance terms on their own (not shown), for children with 2002 remittances observed. The P-value of the lead tests the drought lead in Col. 1 and the triple difference lead in Col. 2. The effect for remittance households is the sum of the two 2009 drought terms. All regressions include child and interview year-month fixed effects, and standard errors are clustered by 2005 municipality. $^{*}p < .10$, $^{**}p < .05$, $^{***}p < .01$.

Table 27—Continuous DID by Subgroup, Height-for-Age

	Change in the height-for-age z-score (HAZ)			
	Boys (1)	Girls (2)	Rural (3)	Urban (4)
Any 2009 deficit	-0.178	-0.234**	0.086	-0.288***
	(0.112)	(0.107)	(0.192)	(0.085)
	[0.132]	[0.036]	[0.659]	[0.007]
Dry vs near normal, $z \leq -1$	-0.005	-0.004	0.170	-0.118
	(0.113)	(0.112)	(0.177)	(0.103)
	[0.965]	[0.972]	[0.390]	[0.289]
Severely dry or worse vs near normal, $z \leq -1.5$	-0.179	-0.120	0.003	-0.243*
	(0.151)	(0.127)	(0.257)	(0.112)
	[0.278]	[0.410]	[0.992]	[0.057]
Dry × 2002 remittances	-0.278	0.119	-0.155	-0.099
	(0.222)	(0.173)	(0.227)	(0.168)
	[0.225]	[0.506]	[0.506]	[0.564]
Children	1,208	1,228	1,020	1,416
Municipalities with no 2009 deficit	30	30	16	20

Notes: This table repeats the continuous DID and SPI results for the WHO height-for-age z-score (HAZ) by subgroup. Cols. 1–2 test for gender disparities in the impact of the drought (2005 roster sex), and Cols. 3–4 test whether it hits rural children harder, as rural household incomes depend more directly on rain-fed agriculture. Rural is a locality with fewer than 2,500 inhabitants (INEGI, 2024). Rows, samples (aged 15 or younger in both rounds) and definitions are as in the main continuous DID and SPI tables, and subgroups with few dry or untreated municipalities give noisy estimates. The any deficit row uses the contdid package, and the SPI rows use the discrete-dose regression in equation 4.1 (Callaway et al., 2024). Standard errors clustered by 2005 municipality are in parentheses. Wild cluster bootstrap p-values are in brackets, which tend to be conservative with few treated municipalities (MacKinnon et al., 2023). Stars use the clustered p-values, and where a wild bootstrap p-value is shown both must pass: $^{*}p < .10$, $^{**}p < .05$, $^{***}p < .01$.

Table 28—Continuous DID by Subgroup, Weight-for-Age

	Change in the weight-for-age z-score (WAZ)			
	Boys (1)	Girls (2)	Rural (3)	Urban (4)
Any 2009 deficit	-0.064	-0.019	0.015	0.016
	(0.155)	(0.108)	(0.243)	(0.090)
	[0.692]	[0.855]	[0.952]	[0.861]
Dry vs near normal, $z \leq -1$	-0.212	-0.210*	-0.133	-0.224**
	(0.170)	(0.107)	(0.209)	(0.104)
	[0.241]	[0.058]	[0.612]	[0.043]
Severely dry or worse vs near normal, $z \leq -1.5$	-0.416	-0.217	-0.381	-0.262*
	(0.244)	(0.163)	(0.311)	(0.139)
	[0.128]	[0.235]	[0.456]	[0.099]
Dry × 2002 remittances	0.063	-0.309	0.084	-0.411
	(0.298)	(0.275)	(0.346)	(0.245)
	[0.842]	[0.263]	[0.825]	[0.100]
Children	556	528	424	660
Municipalities with no 2009 deficit	26	23	12	19

Notes: This table repeats the continuous DID and SPI results for the WHO weight-for-age z-score (WAZ) by subgroup. Cols. 1–2 test for gender disparities in the impact of the drought (2005 roster sex), and Cols. 3–4 test whether it hits rural children harder, as rural household incomes depend more directly on rain-fed agriculture. Rural is a locality with fewer than 2,500 inhabitants (INEGI, 2024). Rows, samples (aged 10 or younger in both rounds) and definitions are as in the main continuous DID and SPI tables, and subgroups with few dry or untreated municipalities give noisy estimates. The any deficit row uses the contdid package, and the SPI rows use the discrete-dose regression in equation 4.1 (Callaway et al., 2024). Standard errors clustered by 2005 municipality are in parentheses. Wild cluster bootstrap p-values are in brackets, which tend to be conservative with few treated municipalities (MacKinnon et al., 2023). Stars use the clustered p-values, and where a wild bootstrap p-value is shown both must pass: $^{*}p < .10$, $^{**}p < .05$, $^{***}p < .01$.

Table 29—Continuous DID by Subgroup, BMI-for-Age

	Change in the BMI-for-age z-score (BMIZ)			
	Boys (1)	Girls (2)	Rural (3)	Urban (4)
Any 2009 deficit	-0.009	-0.043	0.023	-0.049
	(0.115)	(0.090)	(0.120)	(0.103)
	[0.935]	[0.670]	[0.858]	[0.651]
Dry vs near normal, $z \leq -1$	-0.171	-0.085	-0.085	-0.160
	(0.101)	(0.081)	(0.102)	(0.107)
	[0.105]	[0.309]	[0.439]	[0.170]
Severely dry or worse vs near normal, $z \leq -1.5$	-0.144	-0.114	-0.190	-0.087
	(0.100)	(0.090)	(0.099)	(0.079)
	[0.171]	[0.239]	[0.125]	[0.292]
Dry × 2002 remittances	0.541	-0.388**	0.058	0.065
	(0.308)	(0.169)	(0.241)	(0.295)
	[0.107]	[0.030]	[0.824]	[0.842]
Children	1,177	1,206	1,002	1,381
Municipalities with no 2009 deficit	30	30	16	20

Notes: This table repeats the continuous DID and SPI results for the WHO BMI-for-age z-score (BMIZ) by subgroup. Cols. 1–2 test for gender disparities in the impact of the drought (2005 roster sex), and Cols. 3–4 test whether it hits rural children harder, as rural household incomes depend more directly on rain-fed agriculture. Rural is a locality with fewer than 2,500 inhabitants (INEGI, 2024). Rows, samples (aged 15 or younger in both rounds) and definitions are as in the main continuous DID and SPI tables, and subgroups with few dry or untreated municipalities give noisy estimates. The any deficit row uses the contdid package, and the SPI rows use the discrete-dose regression in equation 4.1 (Callaway et al., 2024). Standard errors clustered by 2005 municipality are in parentheses. Wild cluster bootstrap p-values are in brackets, which tend to be conservative with few treated municipalities (MacKinnon et al., 2023). Stars use the clustered p-values, and where a wild bootstrap p-value is shown both must pass: $^{*}p < .10$, $^{**}p < .05$, $^{***}p < .01$.